Enhancing Operational Efficiency and Workload Management in SakhiBoutique – A Case Study of a Home-Based Tailoring Business

A Mid Term report for the BDM capstone Project

Submitted by

Name:Neha Pankaj Mahagaonkar

Roll number:23f1001455

IITM Online BS Degree Program,

Indian Institute of Technology, Madras, Chennai,Tamil Nadu, India, 600036

Contents

1	Executive Summary	4
2	Proof of Originality	4
3	Meta Data	5
4	Descriptive Statistics	6
4	Detailed Explanation of Analysis Process/method	8
5	Results and Findings	10

Declaration Statement

I am working on a Project titled **"Enhancing Operational Efficiency and Workload Management in Sakhi Boutique – A Case Study of a Home-Based Tailoring Business"**. I extend my appreciation to **Sakhi Boutique**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

NehaPankaj

Signature of Candidate:

Name: Neha Pankaj Mahagaonkar

Date: 2and November 2025

1 Executive Summary

Sakhi Boutique is a small,home-based tailoring business located in Shreya Nagar, Osmanpura,Chh Sambhaji Nagar,Maharashtra. Founded by Mrs Rohini Mahagaonkar in year 2006, the boutique operates on B2C model and specializes in Maharashtrian wear such as Saree blouse,kurtis, Salwar suits, Traditional Frocks and dresses, Catering to women in the local community

The boutique currently has 3-4 Employees and generates annual income of approximately Rs 2.5 Lakh. Despite the steady demand the business currently faces challenges related to unstructured operations and irregular employee availability. Manual record keeping for orders, customer measurements and payments results in limited visibility into income and workload distribution . During festive seasons, high demand combined with unpredictable staff attendance leads to delivery delays , increased stress and loss of potential customers

This project aims to enhance operational efficiency by introducing a semi-structured data recording system to record daily operations , monitor workloads, and track payments in an organized manner . the proposed approach will include identification of workflow bottlenecks ,studying demand variations and developing strategies for better workload distribution and scheduling .the expected outcome includes improved productivity, effective task management during peak seasons, and higher profitability supported by structured and data driven decision making

2 Proof Of Originality

As the part of proof of originality for the collected data. I am providing the following supporting documents and materials:Letters from organization, images of boutique,some images of raw data (notebook of owner)and video interaction with the boutique owner with the transcript

BDM Project Proof Of Originality

The mode of video is in marathi language and an english transcript is provided in the google drive above .

This is the Unlisted Youtube Video with subtitles:interaction on youtube(unlisted)

3 Metadata and Descriptive statistics

For this Project the data was systematically collected from the Sakhi Boutique between 1st May 2025 to 16th Oct 2025, the dataset records every order received , detailed information related to the customer item specification cost components and assigned tailor

Each record represented a unique order, enabling comprehensive analysis of cost efficiency, tailor performance and customer demand. The data was imported into Google sheets for cleaning and visualization before statistical analysis using spreadsheet formulas and charts

A	B	C	D	E	F	G	H	I	J	K	L	M
order id	order_date	exp_delivery	delivery date	customer id	item_id	quantity	material_cost	labour_cost	total_cost	net_earning	tailor_name	order status
O1	02/05/2025	07/05/2025	07/05/2025	C12	I1	1	60	80	250	110	sheetal	Completed
O2	02/05/2025	09/05/2025	10/05/2025	C47	I1	2	120	160	500	220	sheetal	Completed
O3	02/05/2025	04/05/2025	10/05/2025	C3	I1	4	240	320	1000	440	sheetal	Completed
O4	03/05/2025	09/05/2025	13/05/2025	C85	I1	2	120	160	500	220	sheetal	Completed
O5	04/05/2025	09/05/2025	09/05/2025	C19	I1	3	180	240	750	330	sheetal	Completed
O6	10/05/2025	15/05/2025	15/05/2025	C33	I1	4	240	320	1000	440	sheetal	Completed

The structured data set comprises the following key fields

1. **Order id** : unique order id for each customer order to ensure tracking.
2. **Order date** : date when customer placed order.
3. **Expected delivery date**: the expected date of delivery.
4. **Delivery date** : the actual delivery date used for analysing lead time and delay time .
5. **Customer id**: pseudomized customer reference,customer names in customer sheet.
6. **Item id** : the product type (eg. , saree blouse, kurti, frock shirt)for order classification.
7. **Quantity**: number of pieces order
8. **Material cost**: total raw material expense
9. **Labor cost**: wage to stitching and service effort
10. **Total cost** : combined cost per order, reflecting total revenue per transaction
11. **Tailor name**: name of service professional responsible for the order
12. **Order status**: completed or cancelled

New Columns created :

13. **Weekstart**: New column created by me for segregating data week wise
14. **Net Earning** : new column for calculating net earning after deducting labor and material cost
15. **Delay days**:New column the number of days the order got delayed
16. **IsWeekend**: New column checking if its a weekend or not
17. **Days to complete**: number of days it takes to complete the order

After cleaning and organizing , the data was analysed in google sheet using pivot tables, descriptive statistics and charts to analyse the sales, cost structure and tailor workload.

Link of dataset : Sakhi boutique orders data

4. Descriptive statistics

4.1 Measure of central tendency

These measures provide insights into general trend of order quantities and revenue

Metric	Quantity	Material cost(Rs.)	Labor cost(Rs.)	Total cost(Rs.)	Net earning(Rs)
Mean	3.06	315.569620	388.03	1252.1518	548.544
Median	3	285	360	1200	440
Mode	3	300	600	1500	440

Interpretation:The average order quality(mean=3.06) indicated that most customers place small to medium sized orders.The mean total cost(Rs.1252.15)suggests typical tailoring jobs fall in the mid-range price categoryThe median and mode values being close to the mean show a consistent pricing pattern with limited skewness.The average net earning (Rs.548.54) highlights a healthy profit margin per order.

4.2 Measure of Variability:

These measures show the degree of variation in order values:

Metric	quantity	Total cost(Rs.)	Net earning(Rs.)
Standard deviation	1.3098	797.01	377.8295
range	1-5	120-3750	40-1750

interpretation : The standard deviation of Rs.797.01 in total cost shows moderate variation in order values– some customers place significantly higher value orders.Net earning Rs.377.83 also vary, reflecting differences in order complexityThe range(Rs.120-Rs.3750) indicated a wide range of service, from simple repairs to premium tailoring projects .

4.3 Frequency analysis

The frequency analysis summarizes the performance of each tailor based on the number of orders handled, avg total cost and highest value transaction.

Tailor	No of Orders	Avg Total Cost(Rs)	Highest Transaction(Rs)
Sheetal	64	1207.8125	3000
Vaishali	51	1451.7464	3750
Shehnaj	43	1081.395349	2250

Interpretation:

1.Vaishali recorded the highest average total cost(Rs.1451.76) per order. This suggests that she often worked in higher value or more complex tailoring tasks, such as designer blouses, gowns etc.

2.Sheetal and Shehnaj gave relatively lower average total cost (1207.81 and 1081.40 respectively), implying they may focus more on regular or lower cost tailoring services.

Overall insights:Sheetal contributes most to order volume, ensuring steady flow of business.Vaishali contributes higher revenue per order, making her important for profitability.Shehnaj maintains moderate performance, balancing both volume and value

5.Methods of Analysis with Justification

5.1Data cleaning and Consolidation

Process:

The Sakhi boutique's order and revenue data was collected from manual registers and digital records maintained over 5 months. The raw data contained inconsistencies such as missing order dates, incomplete delivery details and varying service naming etc . These issues were identified and corrected during the cleaning phase. After cleaning the data daily summaries were generated to represent the number of ordered, total revenue and average costs per day or week

Justification:

Data cleaning ensures that the subsequent analysis reflects accurate business realities. For a small-scale service business, inconsistencies can distort revenue tracking and cost estimation. Consolidation of daily orders provides a unified view of performance trends, allowing

management to identify busy days, estimate revenue inflows, and monitor production load efficiently.

5.2Correlation Analysis(Tailor Assignment and delivery delays)

Process:

A New column, delay_days was Computed as the difference between actual delivery day and expected delivery date. Each order was tagged with the tailor responsible for its completion. Using the CORREL() function in google sheets , correlation was calculated between the number of orders handled by each tailor and their respective average delivery delay days. Additionally, AVERAGEIF() was used to find average completion times of each tailor. done in the tailors summary sheet in the main datasheet.

Justification:

This analysis quantifies the relationship between staff workload and delivery efficiency. By identifying tailors whose assignments show strong positive correlation with higher delay days, management can detect potential inefficiencies caused by overloading or lack of punctuality. These insights enable targeted training, better workload distribution, and improved customer satisfaction through timely service delivery.

5.3 Time Series and Trend Analysis (Revenue and Seasonal Demand)

Process:

Daily and weekly order data were grouped and visualized using pivot tables and line charts. Key metrics such as total orders, total revenue, and 7-day moving averages were plotted to observe fluctuations over time. The FORECAST.LINEAR() function and trendlines were applied to project future order volumes. Additional columns were introduced to mark weekends and festival periods, allowing comparison of demand patterns across time.

Justification:

Trend analysis helps uncover seasonality and demand surges—such as spikes during festivals or weekends. By identifying these high-demand periods, the tailoring shop can plan labour availability, material procurement, and scheduling in advance. Forecasting also supports proactive decision-making, ensuring resource balance and reducing lost revenue opportunities during busy seasons.

5.4 Regression Analysis (Factors Affecting Total Cost)

Process:

Regression analysis was conducted to study how *material cost*, *labour cost*, and *completion time* impact *total cost*. Using built-in formulas such as SLOPE() and INTERCEPT(), linear relationships were established between independent variables (material and labour costs) and the dependent variable (total cost). Scatter plots with trendlines and R^2 values were generated to visualize the strength of these relationships.

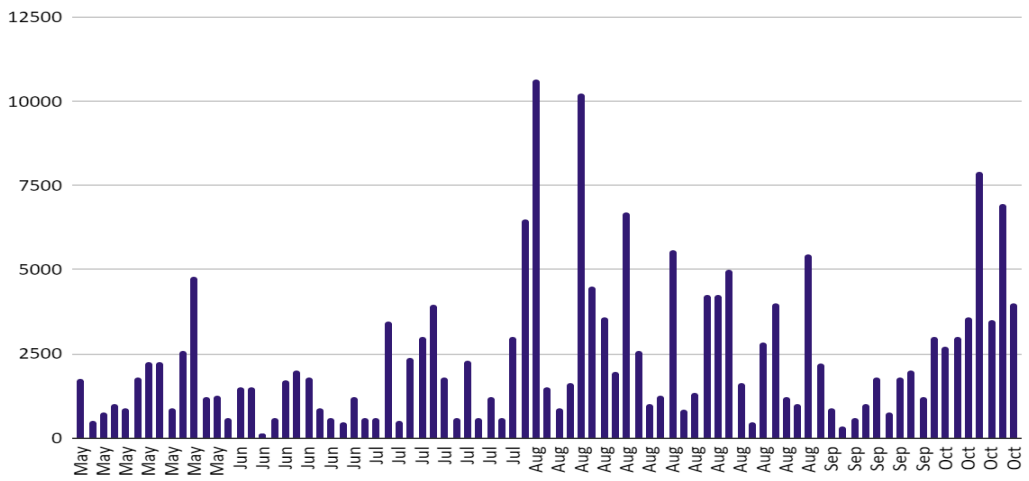
Justification:

Regression analysis quantifies cost dynamics within the production process. Understanding how material or labour fluctuations influence total cost enables management to identify cost drivers and optimize resource allocation. For instance, if labour cost shows a stronger influence on total cost, efforts can focus on productivity improvements or better scheduling to enhance profitability.

6.Results and Findings

6.1 Monthly revenue distribution

The analysis of operational data collected from sakhi boutique provides key insights into the relationship between order volumes staff availability and revenue generation throughout the span of 5 months



This is the barchart of revenue over time, with the x axis showing months(may to october) and y axis shows revenue values up to Rs.12500

The revenue data shows significant fluctuations within each month, with many low values but occasional sharp spikes. This pattern suggests uneven daily sales, possibly due to certain days having bulk or high value orders.

August stands out as the peak month with revenues crossing Rs 10,000 while October also shows noticeable increase performance. In contrast the moths from may to july generally reflect lower revenue level indicating a slower business period during that time

These findings confirm that the boutique's income is highly fluctuated , which directly affects cash flow planning and resource allocation.

6.2Delivery Efficiency Analysis:

The correlation analysis was conducted between the number of tailor assignments and delivery delays

tailor	orderes handled	avg delay
sheetal	64	0.59375
shehnaj	43	0.4418604651
vaishali	51	0.4509803922

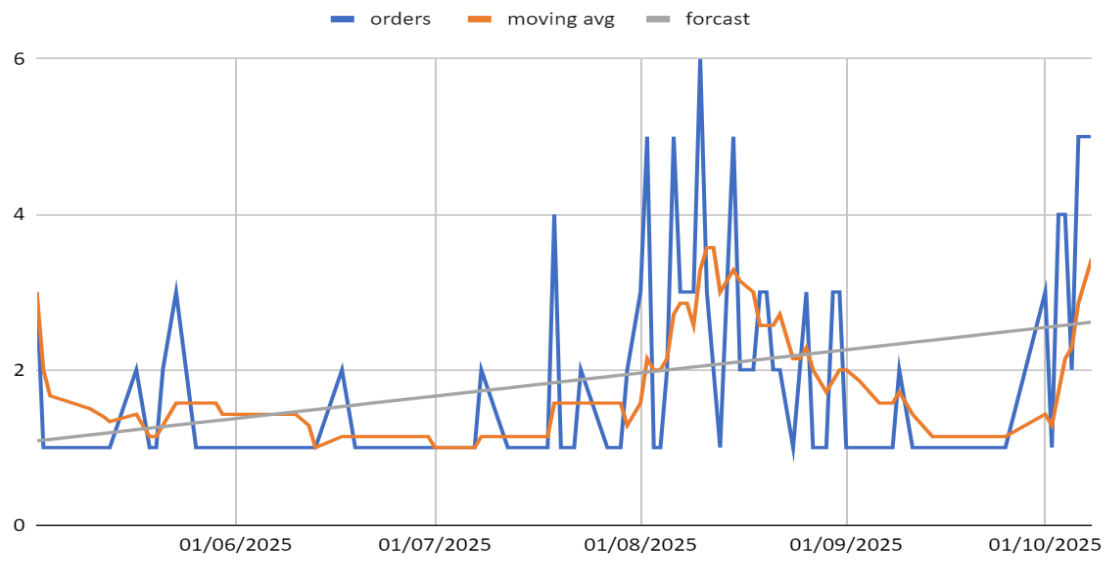
This graph plots Orders Handled on X axis and Average delivery delays on Y axis. It visually represents how the workload of each tailor relates to their delivery perform

ance .The correlation coefficient was found to be 0.9902 indicating that very strong positive . the trendline slopes upwards, it indicates that tailors handling more assignments tend to take longer to deliver. This means that as the workload increases, the delivery delay also rises, suggesting that overload impacts efficiency

From the data Sheetal handled the highest number of orders(64) and recorded the highest avg delay of 0.59 days. In contrast , Shehnaj who handled fewest orders(43), had lowest avg delays of 0.44 days This strongly validates the second problem statement

6.3Time series analysis:

The time series analysis revealed meaningful insights into Sakhi boutiques order and revenue pattern over time.the chart include three main components the blue line representing the actual number of orders placed, the orange line represents 7 day moving average, and the grey line represents forecast trend using FORCAST.LINEAR() function.



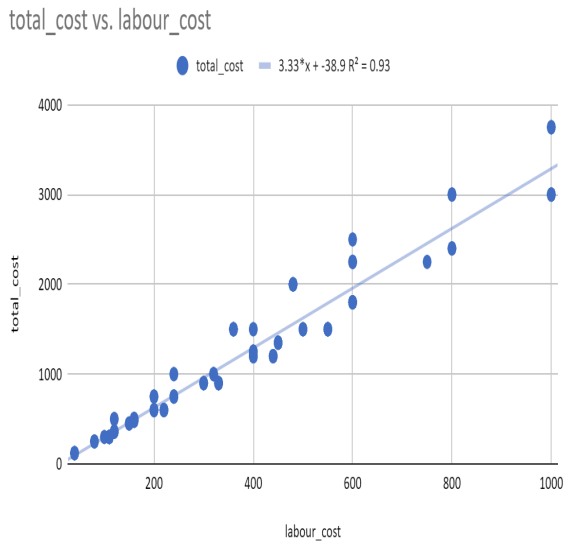
The grey forecast line showed a slight upward slope,indicating a gradual increase in order volumes over time this upward trend reflects positive business momentum.

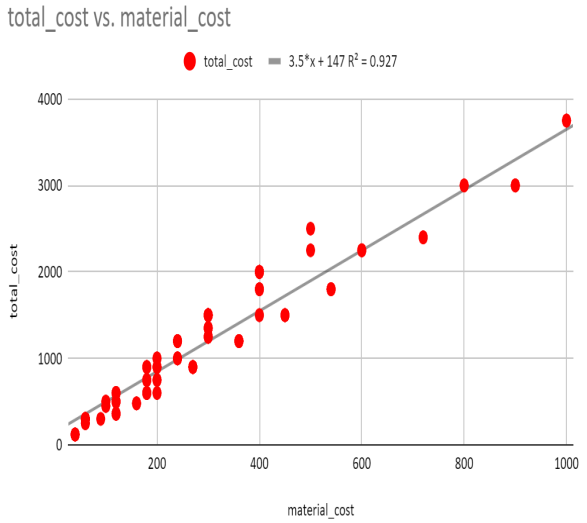
The blue line, which captures daily orders, displayed frequent spikes and drops.these fluctuations indicate volatile daily demand .this pattern indicated strong impact of seasonal and temporal factors on customer orders.

The orange moving average line helped smooth out daily fluctuations, revealing a more consistent underlying trend . from around june to august 20205, the moving average began to rise and stabilize showing that average order volume was gradually increasing during this period

The alignment of the grey forecast line with the orange trend line suggests that the forecasting model accurately captures future demand direction. This provides a reliable basis for anticipating order growth and planning accordingly**.**

6.4Regression analysis:





The regression between material cost and total cost showed an R^2 of 0.99, indicating a near-perfect linear relationship. This means that fluctuations in material prices directly and proportionally affect overall production costs. Since fabric and raw materials are procured regularly in bulk, even small increases in unit price can significantly raise total expenditure.

Similarly, labour cost demonstrated a strong positive correlation ($R^2 = 0.98$), highlighting its substantial impact on total cost. This suggests that higher labour intensity or inefficiencies in tailoring hours directly contribute to increased production expenses. As labour accounts for a major share of operational costs, optimizing task allocation and tracking productivity can meaningfully improve profit margins.

Overall, these results confirm that both material and labour costs are primary cost drivers in the tailoring process. Strengthening cost monitoring, negotiating better supplier terms, and improving workforce scheduling could help the boutique achieve greater cost efficiency, maintain profitability, and better manage seasonal fluctuations in demand.